# Appendix

## A Training Details

For wide ResNets, we look at models with depths of 14, 20, 26, and 38, and width multipliers of 1, 2, 4, 8 and 10. For deep ResNets, we experiment with depths 32, 44, 56, 110 and 164. In CIFAR-100 experiments, the block structure only appears at a greater depth so we also include depths 218 and 224 in our investigation. For Patch Camelyon datasets, we find that depth 80 is enough to induce a block structure in the internal representations. All ResNets follow the architecture design in [12, 41].

Unless otherwise specified, we train all the models using SGD with momentum of 0.9 for 300 epochs, together with a cosine decay learning rate schedule and batch size of 128. Learning rate is tuned with values [0.005, 0.01, 0.001] and $L_2$ regularization strength with values [0.001, 0.005]. For CIFAR-10 and CIFAR-100 experiments, we apply standard CIFAR-10 data augmentation consisting of random flips and translations of up to 4 pixels. With Patch Camelyon, we use random crops of size 32x32, together with random flips, to obtain the training data. At test time, the networks are evaluated on central crops of the original images. For CKA analysis, each architecture is trained with 10 different seeds and evaluated on the full test set of the corresponding domain.

# B Block Structure on Out-of-Distribution Data

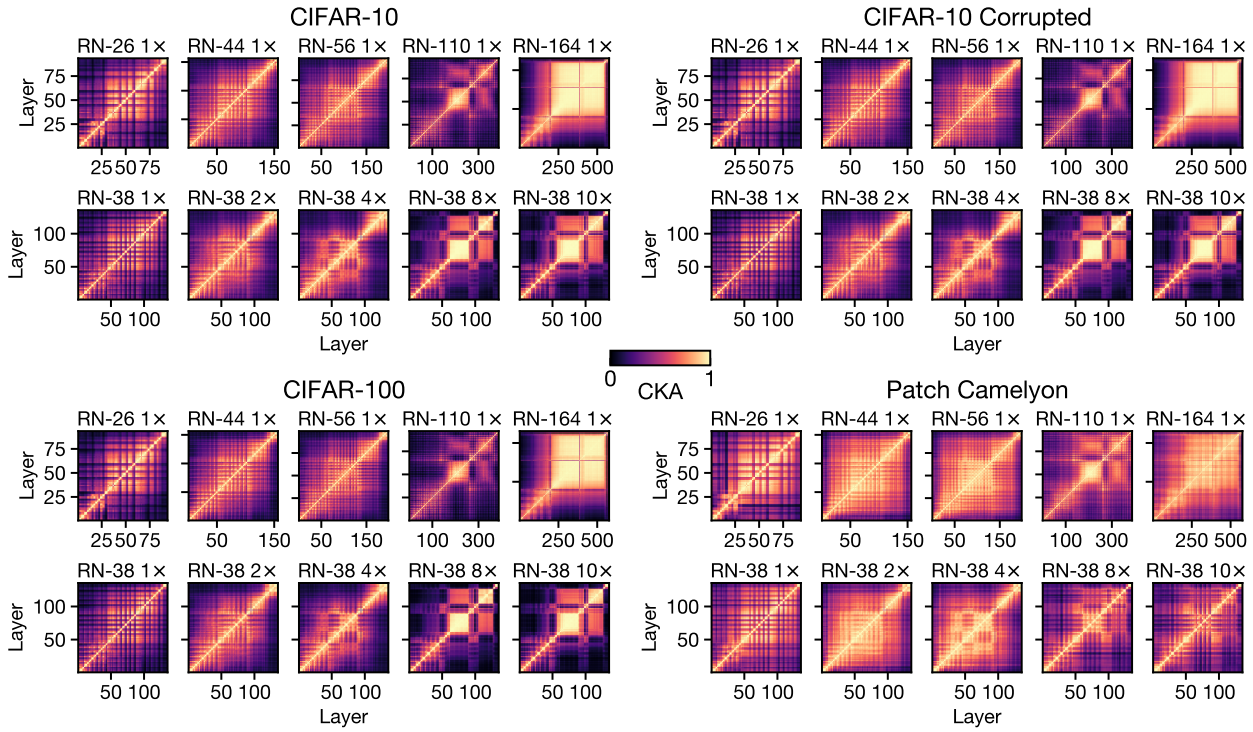


**Figure B.1: Appearance of block structure depends on the data on which representations are computed.** We plot CKA heatmaps for models of varying depths (top rows) and widths (bottom rows) trained on CIFAR-10, evaluated on different datasets ordered by the degree of out-of-distribution. We observe that the block structure representation are robust to small distribution shifts in the data, as evident from CKAs computed on CIFAR-10 corrupted dataset (which adds perturbations to the original CIFAR-10 data) and CIFAR-100 dataset (which contains mutually exclusive classes but undergoes the same data collection procedure as CIFAR-10). However, larger shifts, such as from CIFAR-10 to Patch Camelyon, produce significantly different representational structures.

# C Block Structure and the First Principal Component

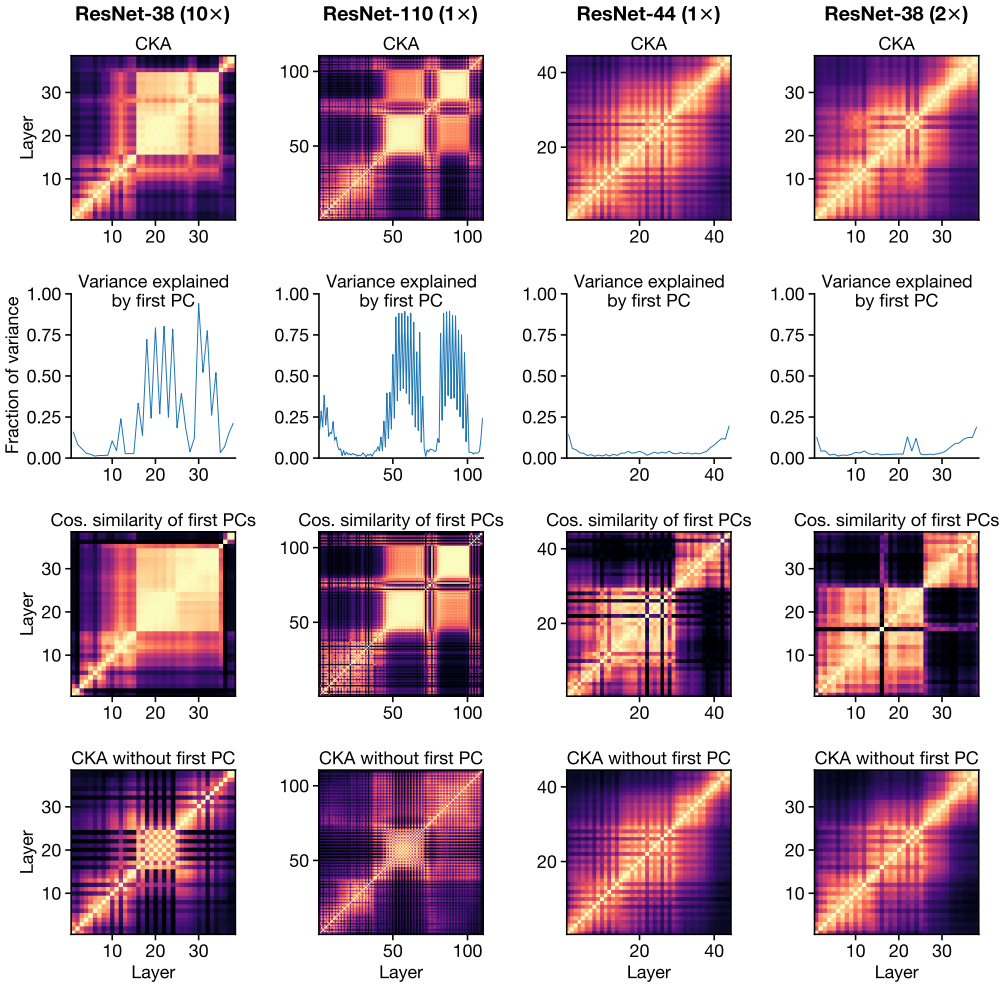


**Figure C.1: The relationship between block structure and the first principal component**. Each column represents a different architecture. In ResNet-110 (1×) and ResNet-38 (10×), we observe a block structure in the CKA plot (top row), and find that the first principal component explains a large fraction of the variance in the layers that comprise the block structure (second row). We also observe that the cosine similarity of the first PCs (third row) resembles the CKA plot, and removing the first PC before computing CKA substantially attenuates the block structure (bottom row). By contrast, in ResNet-44 (1×) and ResNet-38 (2×), which have no block structure, the first PC explains only a small fraction of the variance, and the CKA plot does not resemble the cosine similarity between the first PCs, but instead resembles CKA computed without the first PCs.